

MLOps – Assignment : 01

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Objective :

The objective of this assignment is to implement data processing using Apache Pig on a Hadoop environment. **The assignment performs the following tasks:**

- Start the Hadoop HDFS services.
- Start the Hadoop YARN services.
- Verify that the Hadoop daemons are running.
- Store a CSV dataset in HDFS.
- Read and process the dataset using an Apache Pig script.
- Calculate the average rating for each product category.
- Store the processed result in HDFS.
- Verify and display the generated output.

Dataset :

The input dataset is a CSV file named reviews.csv stored in HDFS at:

- `/user/aiml/reviews/reviews.csv`

The dataset contains three fields:

- **ID** – Unique identifier for each review.
- **Category** – Product category.
- **Rating** – Rating given to the product.

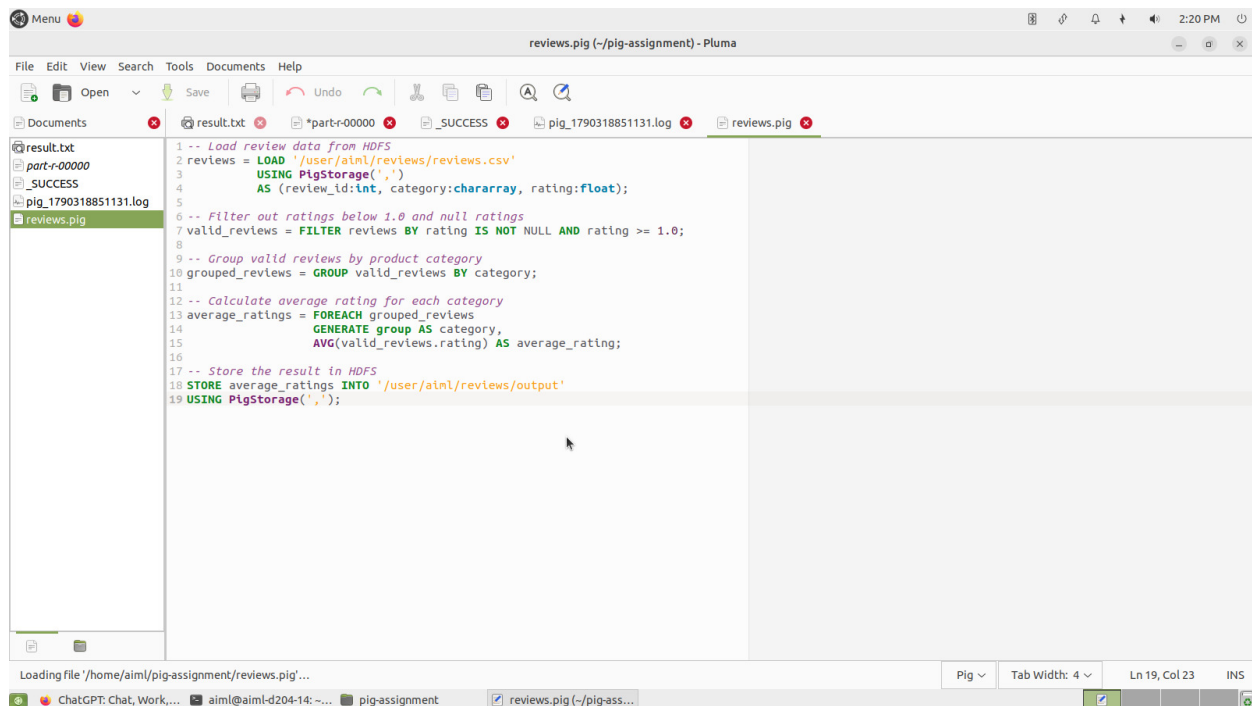
INPUT:

- 1,Electronics,4.5
- 2,Electronics,3.0
- 3,Electronics,0.5
- 4,Clothing,5.0
- 5,Clothing,4.0

- 6,Clothing,0.0
- 7,Books,4.0
- 8,Books,3.5
- 9,Books,
- 10,Home,2.5

Pig Script :

The Apache Pig program loads the review dataset from HDFS, filters out invalid ratings, groups the remaining records by product category, calculates the average rating for each category, and stores the final result back in HDFS.



```

1 -- Load review data from HDFS
2 reviews = LOAD '/user/aiml/reviews/reviews.csv'
3             USING PigStorage(',')
4             AS (review_id:int, category:chararray, rating:float);
5
6 -- Filter out ratings below 1.0 and null ratings
7 valid_reviews = FILTER reviews BY rating IS NOT NULL AND rating >= 1.0;
8
9 -- Group valid reviews by product category
10 grouped_reviews = GROUP valid_reviews BY category;
11
12 -- Calculate average rating for each category
13 average_ratings = FOREACH grouped_reviews
14                   GENERATE group AS category,
15                             AVG(valid_reviews.rating) AS average_rating;
16
17 -- Store the result in HDFS
18 STORE average_ratings INTO '/user/aiml/reviews/output'
19 USING PigStorage(',');
  
```

Code Explantion :

The **LOAD** operation is used to read the reviews.csv dataset from HDFS. The PigStorage(',') function specifies that the fields in the input file are separated by commas. The schema defines review_id as an integer, category as a character string, and rating as a floating-point value.

The **FILTER** operation removes invalid records from the dataset. The condition rating IS NOT NULL AND rating >= 1.0 ensures that records with missing ratings and ratings below 1.0 are excluded from further processing. Therefore, ratings such as 0.5 and 0.0 are removed, while the missing rating in the Books category is also excluded.

The **GROUP** operation groups the valid records according to their product category. After grouping, all valid Electronics reviews are placed into one group, Clothing reviews into another group, and similarly for Books and Home.

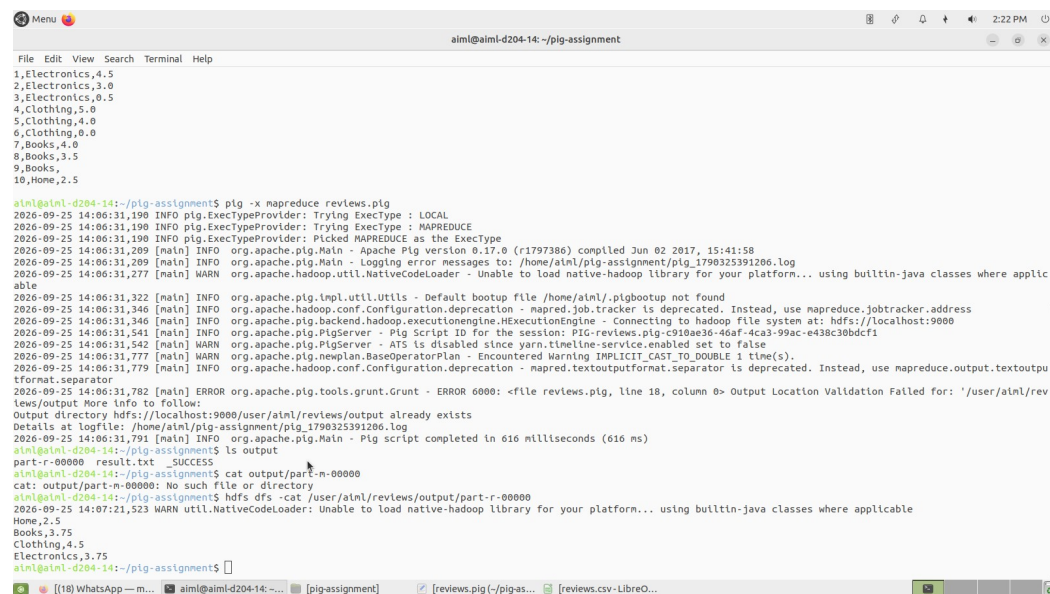
The **FOREACH** operation is then used to process each category group. The `AVG()` function calculates the average rating of the valid reviews within each category. The category name and its calculated average rating are then generated as the final result.

Finally, the **STORE** operation saves the calculated category-wise average ratings in the HDFS directory `/user/aiml/reviews/output`. The `PigStorage(',')` function stores the output fields in comma-separated format.

Execution :

The Hadoop environment was started and the Pig program was executed in MapReduce mode using the following main commands:

- `start-dfs.sh`
- `jps`
- `start-yarn.sh`
- `jps`
- `hdfs dfs -cat /user/aiml/reviews/reviews.csv`
- `pig -x mapreduce reviews.pig`
- `hdfs dfs -cat /user/aiml/reviews/output/part-r-00000`



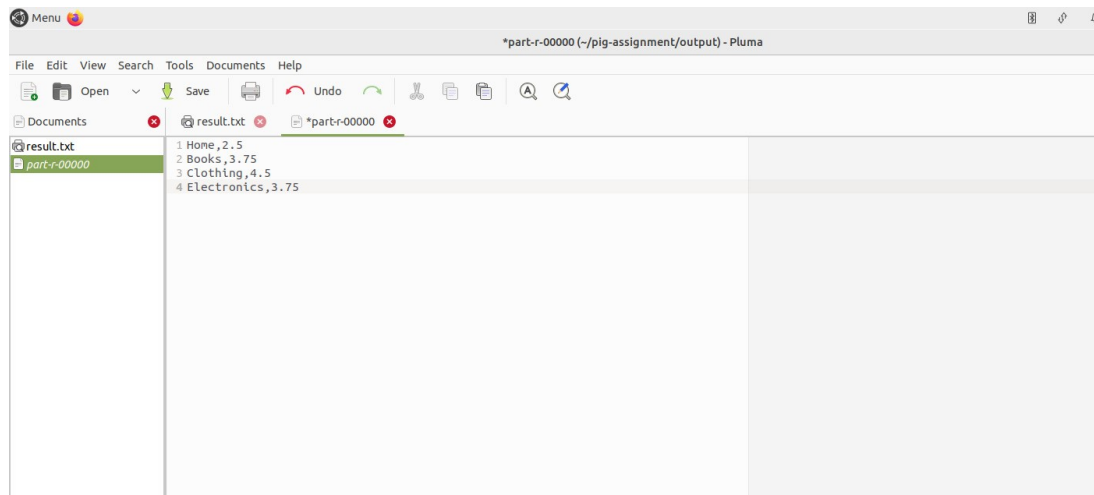
```
aiml@aiml-d204-14: ~/pig-assignment
File Edit View Search Terminal Help
1,Electronics,4.5
2,Electronics,3.0
3,Electronics,0.5
4,Clothing,5.0
5,Clothing,4.0
6,Clothing,0.0
7,Books,4.0
8,Books,3.5
9,Books,
10,Home,2.5

aiml@aiml-d204-14:~/pig-assignment$ pig -x mapreduce reviews.pig
2026-09-25 14:06:31,190 INFO pig.ExecTypeProvider: Trying ExecType : LOCAL
2026-09-25 14:06:31,190 INFO pig.ExecTypeProvider: Trying ExecType : MAPREDUCE
2026-09-25 14:06:31,190 INFO pig.ExecTypeProvider: Picked MAPREDUCE as the ExecType
2026-09-25 14:06:31,209 [main] INFO org.apache.pig.Main - Apache Pig version 0.17.0 (r1797386) compiled Jun 02 2017, 15:41:58
2026-09-25 14:06:31,209 [main] INFO org.apache.pig.Main - Logging error messages to: /home/aiml/pig-assignment/pig_1798325391286.log
2026-09-25 14:06:31,277 [main] WARN org.apache.hadoop.util.NativeCodeLoader - Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
2026-09-25 14:06:31,322 [main] INFO org.apache.pig.impl.util.Utils - Default bootstrap file /home/aiml/.pigbootstrap not found
2026-09-25 14:06:31,346 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.job.tracker is deprecated. Instead, use mapreduce.jobtracker.address
2026-09-25 14:06:31,346 [main] INFO org.apache.pig.backend.hadoop.executionengine.HExecutionEngine - Connecting to hadoop file system at: hdfs://localhost:9000
2026-09-25 14:06:31,541 [main] INFO org.apache.pig.PigServer - Pig Script ID for the session: PIG-reviews.pig-c910ae36-46af-4ca3-99ac-e438c30bdcf1
2026-09-25 14:06:31,542 [main] WARN org.apache.pig.PigServer - ATS is disabled since yarn.timeline.service.enabled set to false
2026-09-25 14:06:31,777 [main] WARN org.apache.pig.nemPlan.BaseOperatorPlan - Encountered warning IMPLICIT_CAST_TO_DOUBLE 1 time(s).
2026-09-25 14:06:31,779 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.textoutputformat.separator is deprecated. Instead, use mapreduce.output.textoutputformat.separator
2026-09-25 14:06:31,782 [main] ERROR org.apache.pig.tools.grunt.Grunt - ERROR 6000: <file reviews.pig, line 18, column 0> Output Location Validation Failed for: '/user/aiml/reviews/output' More info to follow:
Output directory hdfs://localhost:9000/user/aiml/reviews/output already exists
Details at logfile: /home/aiml/pig-assignment/pig_1798325391286.log
2026-09-25 14:06:31,791 [main] INFO org.apache.pig.Main - Pig script completed in 616 milliseconds (616 ns)
aiml@aiml-d204-14:~/pig-assignment$ ls output
part-r-00000 result.txt _SUCCESS
aiml@aiml-d204-14:~/pig-assignment$ cat output/part-r-00000
cat: output/part-r-00000: No such file or directory
aiml@aiml-d204-14:~/pig-assignment$ hdfs dfs -cat /user/aiml/reviews/output/part-r-00000
2026-09-25 14:07:21,523 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
Home,2.5
Books,3.75
Clothing,4.5
Electronics,3.75
aiml@aiml-d204-14:~/pig-assignment$
```

Output :

After successful execution, the generated output is stored in HDFS at `/user/aiml/reviews/output/part-r-00000`. Since ratings below 1.0 and null ratings are filtered out, the average ratings are calculated only from valid records. The final output is:

The Electronics category has valid ratings of 4.5 and 3.0, giving an average of 3.75 after excluding 0.5. The Clothing category has valid ratings of 5.0 and 4.0, giving an average of 4.5 after excluding 0.0. The Books category has valid ratings of 4.0 and 3.5, while the missing rating is excluded, giving an average of 3.75. The Home category contains a valid rating of 2.5, so its average is 2.5.



Outcome :

The Apache Pig program was successfully executed in MapReduce mode on the Hadoop environment. The review dataset was loaded from HDFS, and invalid records containing ratings below 1.0 or missing ratings were filtered out. The valid records were grouped according to product category, and the average rating for each category was calculated using the AVG() function.

The final category-wise results were successfully stored in HDFS. The experiment demonstrates the use of Apache Pig for data loading, filtering, grouping, aggregation, and storing processed results in a Hadoop environment. The output confirms that the required review data processing was completed successfully.